Oxford Cambridge and RSA Examinations

GCE

Economics

H460/02: Macroeconomics

A Level

Mark Scheme — Predicted Paper 2 (Hard)

MARKING INSTRUCTIONS

Marks awarded must relate directly to the marking criteria. Alternative correct answers and unexpected approaches must be credited where they show relevance. Mark schemes should be read in conjunction with the published question paper.

For answers marked by levels of response:

- To determine the level — start at the highest level and work down until you reach the level that matches the answer.
- To determine the mark within the level, consider whether the answer is at the borderline (bottom), just enough (above bottom/middle), meets criteria with slight inconsistency (above middle), or consistently meets the criteria (top).

Question 1 (a)

Answer	Mark	Guidance
Identify two factors that may have caused the rise in the UK's inflation rate in 2022. Two from: • Rising energy prices (1) • Disrupted supply chains (1) • Tight labour markets (1) • Rising nominal wages / wage-price pressure (1)	2	Annotate with ✔ Do not accept generic factors (e.g. "government policy") not referenced in the stimulus.

Question 1 (b)

Answer	Mark	Guidance
Calculate to one decimal place the UK's terms of trade in 2024. **Correct answer = 94.8**	1	Annotate with ✔ Index of export prices / Index of import prices × 100 118.5 / 125.0 × 100 = 94.80 = 94.8 (to 1 d.p.) Do not credit 0.948 or 95.

Question 1 (c)

Answer	Mark	Guidance
Explain whether the relationship in Fig. 1 between labour productivity growth and real wage growth is the expected one. • Yes — broadly the expected positive relationship (1) • Theory expects higher productivity growth to lead to higher real wage growth because wages tend to reflect the marginal product of labour (1) • Supporting evidence e.g. South Korea has the highest productivity growth (2.1%) and the highest real wage growth (1.8%); the UK and Germany have the lowest productivity growth (0.4% and 0.2%) and negative real wage growth (−0.8% and −0.5%) (1) • Exception: Japan has higher productivity growth (0.9%) than France (0.6%) but lower real wage growth (0.1% vs 0.2%), suggesting that other factors such as labour market institutions, union bargaining power and inflation expectations also affect real wages (1)	3	Annotate with ✔ Relationship — 1 Why expected — 1 Supporting evidence — 1 Exception — 1

Question 1 (d)

Answer	Mark	Guidance
Using Table 1, explain the relationship between government debt and 10-year bond yields. • There is no strong / consistent relationship between government debt and bond yields (1) • Evidence e.g. Japan has by far the highest government debt (252% of GDP) but the lowest bond yield (1.1%); Italy has lower debt (137%) than Japan but a higher yield of 3.8%; Germany has the lowest debt (64%) and one of the	4	Annotate with ✔ Relationship — 1 Evidence — 1 Explanation — 1 Exception / limitation — 1

lowest yields (2.3%) (1) • Possible explanation: yields reflect the risk of default, expectations of future inflation, monetary policy and the currency of the debt. Japan's debt is largely held domestically and denominated in yen, and the Bank of Japan actively buys bonds, keeping yields low. Italy's higher yield reflects perceived default/redenomination risk within the eurozone (1) • Exception / limitation: the relationship may be visible within a peer group (e.g. eurozone: Germany low debt / low yield; Italy higher debt / higher yield), but cross-country comparisons are misleading because each country's institutional context differs (1)		

Question 1 (e)* Evaluate whether the Bank of England's rise in interest rates is likely to reduce UK inflation without damaging economic growth. [8]

Level / Mark	Descriptor
Level 2 (5–8)	Good KU, good application with data, strong analysis, strong evaluation and a logical supported judgement.
Level 1 (1–4)	Reasonable KU, reasonable application (possibly inconsistent), reasonable analysis in single links, limited evaluation.
0 marks	Response is not worthy of credit

Indicative content — Question 1(e)*

Knowledge and understanding

- Interest rates are the cost of borrowing / reward for saving.
- Higher interest rates are contractionary monetary policy, intended to reduce AD and inflation.
- There may be a short-run trade-off between inflation and output (Phillips curve).

Application — context

- Bank Rate rose from 0.1% in December 2021 to 5.25% in August 2023.
- Inflation peaked at 11.1% in 2022 and fell back to around 4.0% in 2024.
- Real GDP growth was only 0.9% in 2024.

Analysis — rate rises reduce inflation without much growth damage

- Higher rates raise the cost of borrowing for households and firms, reducing consumption and investment (components of AD).
- A rising exchange rate from higher rates makes imports cheaper, directly reducing imported inflation.
- A fall in AD relative to LRAS reduces demand-pull inflation.
- Raising interest rates anchors inflation expectations, reducing second-round effects on wages and prices.
- Since inflation peaked in 2022 and fell to around 4.0% in 2024, the policy has had some apparent success.

Evaluation — rate rises may damage growth

- Higher borrowing costs reduce investment, which lowers long-run productive capacity.
- Real GDP growth of only 0.9% in 2024 is very weak and may reflect the contractionary effect of rate rises.
- Much of the inflation was driven by supply-side shocks (energy prices, supply chains) — monetary policy is less effective against cost-push inflation.
- Higher mortgage costs reduce disposable income of homeowners, further depressing consumption.
- Transmission lags of 12–18 months mean the full effect on growth may not yet be visible.
- The UK's weak productivity growth (0.4%) limits the economy's capacity to absorb demand — so smaller cuts in AD may have bigger output costs.

Judgement may include

- The policy has succeeded in reducing headline inflation but at the cost of weak growth — a trade-off is evident.
- Effectiveness depends on whether inflation is primarily cost-push (monetary policy less effective) or demand-pull.
- The weakness of UK productivity means the supply-side fundamentals need addressing too.
- Time lags mean the full impact on growth may be yet to come.

Question 1 (f)*   Evaluate whether the UK government should prioritise reducing its budget deficit. [12]

Level / Mark	Descriptor
Level 3 (9–12)	Good KU and application, strong analysis (well-developed chains of reasoning), strong evaluation with a logical supported judgement.
Level 2 (5–8)	Good KU and application, reasonable analysis (single links), reasonable evaluation with an unsupported judgement.
Level 1 (1–4)	Reasonable KU, reasonable application, limited analysis and limited evaluation.
0 marks	Response is not worthy of credit

Indicative content — Question 1(f)*

Knowledge and understanding

- A budget deficit arises when government spending exceeds tax revenue in a year.
- Public debt is the accumulated stock of past deficits.
- Reducing a deficit requires some combination of higher taxes and/or lower government spending (contractionary fiscal policy).

Application — context

- UK public sector net debt has reached 98% of GDP.
- Interest payments on debt have more than doubled between 2021 and 2023.
- About a quarter of UK government debt is index-linked, raising servicing costs as inflation rose.

Analysis — UK should prioritise reducing the deficit

- Rising debt-servicing costs divert resources from productive government spending.
- A growing debt-to-GDP ratio increases perceptions of default risk, potentially raising bond yields and further increasing debt-servicing costs.
- Reducing the deficit now may lower long-term interest rates, supporting private sector investment (reduced financial crowding out).
- Controlled fiscal policy may support confidence in the pound and reduce imported inflation.
- With inflation still above target, fiscal tightening complements the BoE's contractionary monetary policy, making it more effective.

Evaluation — UK should not prioritise deficit reduction

- Contractionary fiscal policy risks pushing the economy into recession, given weak growth of only 0.9% in 2024.
- Cutting government investment (infrastructure, education) harms long-run productive capacity and worsens the productivity problem.
- Japan's 252% debt-to-GDP ratio (Table 1) shows that high debt does not automatically translate into high yields or a crisis.
- If fiscal tightening coincides with monetary tightening, the combined effect on AD can be excessive.
- In a recession, the deficit typically falls as the economy recovers — "austerity" may be self-defeating if it causes nominal GDP to fall too.
- Higher taxes reduce disposable income, worsening the cost-of-living squeeze on households.

Judgement may include

- The answer depends on the composition — cutting current spending is less harmful than cutting capital spending.
- The state of the economy matters — weak growth argues against rapid deficit reduction; overheating would argue for it.
- Market conditions matter — if bond yields remain manageable, there is less urgency.

- A credible medium-term plan may be more valuable than rapid short-term cuts.
- Distribution of the adjustment (progressive tax rises vs regressive spending cuts) has welfare implications.

Section B — Question 2*

Evaluate, with the use of an appropriate diagram(s), whether an appreciation of a country's currency will benefit the economy. [25]

Level / Mark	Descriptor
Level 5 (21–25)	Strong KU, application, analysis and evaluation. Accurate, integrated diagram(s). Logical and well-supported judgement.
Level 4 (16–20)	Good KU, application, analysis and evaluation. Predominantly correct diagram(s). Unsupported judgement.
Level 3 (11–15)	Good KU, application, good analysis, reasonable evaluation. Unsupported judgement.
Level 2 (6–10)	Reasonable KU, application, analysis (single links) and evaluation.
Level 1 (1–5)	Limited KU, application, analysis and evaluation.
0 marks	Response is not worthy of credit.

Indicative content — Question 2*

Knowledge and understanding

- Currency appreciation is a rise in the value of one currency relative to others under a floating exchange rate system.
- SPICED: Strong Pound / currency — Imports Cheaper, Exports Dearer.
- Macroeconomic objectives: low stable inflation, growth, employment, current account stability.

Analysis — appreciation benefits the economy

- Cheaper imports reduce cost-push inflation, particularly for energy and commodities.
- Lower import prices raise real disposable income, supporting consumption.
- Cheaper imported capital goods reduce the cost of investment, raising LRAS.
- Appreciation signals a strong economy, attracting portfolio investment.
- Diagram: AS shifts rightward as import costs fall; price level falls, real GDP rises.

Evaluation — appreciation may not benefit the economy

- Exports become more expensive, reducing export volumes and worsening net exports (AD falls).
- Deterioration in the current account may raise unemployment in export-reliant sectors.
- If the Marshall–Lerner condition holds, the current account worsens.
- Lower AD can slow growth and raise unemployment (diagram shows AD shifting left).
- Export-oriented firms may face pressure to cut wages or close, causing structural unemployment.

Possible judgement

- The net effect depends on price elasticities of demand for exports and imports (M-L condition).
- A small, gradual appreciation may be manageable; a large rapid one is destabilising.
- An appreciation may be beneficial when inflation is the main concern but damaging when growth is weak.
- Effect differs between economies — import-dependent economies benefit more from cheaper imports.

Section B — Question 3*

Evaluate, with the use of an appropriate diagram(s), whether cost-push inflation is more harmful than demand-pull inflation. [25]

Level descriptors as for Q2.*

Indicative content — Question 3*

Knowledge and understanding

- Cost-push inflation: rise in price level caused by rise in costs of production (LRAS/SRAS shifts left).
- Demand-pull inflation: rise in price level caused by excess aggregate demand (AD shifts right).
- Both reduce real incomes, though through different mechanisms.

Analysis — cost-push is more harmful

- Cost-push inflation is accompanied by falling real GDP — stagflation — raising unemployment as well as inflation.
- Monetary policy is less effective against cost-push inflation: raising rates to reduce demand worsens unemployment without addressing the cost shock.
- Cost-push inflation is often driven by supply shocks outside policymakers' control (oil prices, war, supply chain disruption).
- Cost-push inflation triggers wage–price spirals if workers demand higher wages to protect real incomes.
- Diagram: SRAS shifts left — price level rises and output falls.

Evaluation — demand-pull may be equally/more harmful

- Demand-pull inflation caused by excessive expansionary policy can spiral out of control if not addressed.
- Persistent demand-pull inflation erodes savings and creates uncertainty that reduces investment.
- Demand-pull inflation can be addressed by contractionary policy; cost-push cannot easily.
- Demand-pull often coincides with strong growth and low unemployment, so welfare effects are mixed.
- High demand-pull inflation that is expected to persist raises nominal interest rates and damages the financial system.

Possible judgement

- Cost-push inflation creates harder trade-offs for policymakers, so may be more harmful.
- The harm of each depends on the cause, persistence and how expectations are anchored.
- Hyperinflation (usually demand-pull / monetary) is the most harmful form of all.
- Cost-push inflation's harm depends on how long supply shocks last.

Section C — Question 4*

Evaluate whether progressive taxation is the most effective way to reduce income inequality. [25]

Level descriptors as for Q2.*

Indicative content — Question 4*

Knowledge and understanding

- Progressive tax: average tax rate rises with income (e.g. UK income tax, inheritance tax).
- Regressive tax: average tax rate falls with income (e.g. VAT tends to be regressive).
- Income inequality is typically measured by the Gini coefficient.

Analysis — progressive taxation reduces inequality

- Higher rates on higher incomes directly redistribute post-tax income.
- Revenue can fund welfare, education and healthcare that disproportionately benefit low-income groups.
- Inheritance tax reduces intergenerational transfer of wealth.
- Reduces the market income gap after accounting for redistribution (post-tax Gini < pre-tax Gini).
- Can fund targeted benefits (tax credits, universal credit) that raise low incomes.

Evaluation — may not be most effective

- High marginal tax rates may reduce work incentives (Laffer curve) — reducing the revenue available for redistribution.
- High earners can avoid tax through legal avoidance, emigration, or shifting income into untaxed forms.
- Other policies may be more effective: minimum wage, education access, trade-union protection.
- Benefits system may be more directly redistributive than taxation (universal credit, child benefit).
- Supply-side measures (education, skills) address root causes rather than symptoms.
- Indirect taxes fund public services even though they are regressive, so they complement progressive direct taxation.

Possible judgement

- A combination of progressive taxation and well-targeted benefits is usually most effective.
- Minimum wages and education address pre-tax inequality directly.
- Effectiveness of progressive tax depends on how leaky the system is (avoidance, evasion).
- Long-run effect depends on how redistributive spending is financed and directed.

Section C — Question 5*

Evaluate whether a country is likely to benefit from joining a monetary union such as the eurozone. [25]

Level descriptors as for Q2.*

Indicative content — Question 5*

Knowledge and understanding

- A monetary union involves members adopting a common currency and ceding control of monetary policy to a shared central bank.
- The eurozone has a single currency (euro), a single central bank (ECB) and a single monetary policy.

Analysis — joining may benefit a country

- Elimination of exchange rate uncertainty within the union encourages trade and FDI.
- Reduction in transaction costs on intra-union trade.
- Greater price transparency increases competition, benefiting consumers.
- Access to deeper, more liquid financial markets lowers borrowing costs.
- Credibility of low inflation from a strong central bank (e.g. ECB's inflation target).

Evaluation — joining may not benefit a country

- Loss of independent monetary policy — "one size fits all" may be inappropriate for individual economies.
- Loss of exchange rate as an adjustment mechanism for asymmetric shocks.
- Fiscal constraints (Stability and Growth Pact) limit expansionary fiscal response.
- Examples: Greece and other periphery economies struggled during 2010–2012 sovereign debt crisis.
- Reduced ability to respond to idiosyncratic shocks (Italy's structural weaknesses, Ireland's banking crisis).
- Political costs — loss of monetary sovereignty.

Possible judgement

- Benefits depend on whether the country is an Optimum Currency Area member with the union (labour mobility, similar business cycles, fiscal transfers).
- Small open economies with close trade links gain more than large diverse economies.
- The design of the monetary union matters — eurozone lacks fiscal union, amplifying costs.
- Long-run benefits of low inflation may outweigh short-run adjustment costs.

ASSESSMENT OBJECTIVES GRID

Question	AO1	AO2	AO3	AO4	Total	Quant.
1(a)		2			2	
1(b)		1(1)			1	(1)
1(c)	2	1(1)			3	(1)
1(d)	2(2)	2(2)			4	(4)
1(e)*	1	1	3	3	8	
1(f)*	1	1	5(2)	5	12	(2)
2*/3*	6(2)	6(2)	6(2)	7(2)	25	(8)
4*/5*	6	6	6	7	25	
TOTAL	**18**	**20**	**20**	**22**	**80**	**(16)**

END OF MARK SCHEME